Hackability is particularly concerning given arguments that reward optimizing behavior tends to be power-seeking (Turner et al., 2021). But Leike et al. (2018) establish that any desired behavior (power-seeking or not) can in principle be specified as optimal via a reward function.[3] However, unlike us, they do not consider the entire policy preference ordering. Meanwhile, Abel et al. (2021) note that Markov reward functions cannot specify arbitrary orderings over policies or trajectories, although they do not consider hackability. Previous works consider reward functions to be equivalent if they preserve the ordering over policies (Ng et al., 1999, 2000). Unhackability relaxes this, allowing equalities to be refined to inequalities, and vice versa. Unhackability provides a notion of what it means to be "aligned enough"; Brown et al. (2020b) provide an alternative. They say a policy is ε-value aligned if its value at every state is close enough to optimal (according to the true reward function). Neither notion implies the other.

Reward tampering (Everitt et al., 2017; Kumar et al., 2020; Uesato et al., 2020; Everitt et al., 2021) can be viewed as a special case of reward hacking, and refers to an agent corrupting the process generating reward signals, e.g. by tampering with sensors, memory registers storing the reward signal, or other hardware. Everitt et al. (2017) introduce the Corrupt Reward MDP (CRMDP), to model this possibility. A CRMDP distinguishes corrupted and uncorrupted rewards; these are exactly analogous to the proxy and true reward discussed in our work and others. Leike et al. (2018) distinguish reward tampering from *reward gaming*, where an agent achieves inappropriately high reward without tampering. However, in principle, a reward function could prohibit all forms of tampering if the effects of tampering are captured in the state. So this distinction is somewhat imprecise, and the CRMDP framework is general enough to cover both forms of hacking.

Our notion of simplification bears a close resemblance to quantilization (Taylor, 2016). Quantilization returns a random policy from the top n% best policies. This is similar to equating the values of those policies, but a simplification may also equate the values of the bottom/middle n%, etc. Thus simplification may achieve a similar effect to quantilization without assuming that we are free to choose from among the best policies.

4 Preliminaries

We begin with an overview of reinforcement learning (RL) to establish our notation and terminology. Section 4.2 introduces our novel definitions of hackability and simplification.

4.1 Reinforcement Learning

We expect readers to be familiar with the basics of RL, which can be found in Sutton and Barto (2018). RL methods attempt to solve a sequential decision problem, typically formalised as a **Markov decision process (MDP)** , which is a tuple $(S, A, T, I, \mathcal{R}, \gamma)$ where S is a set of states, A is a set of actions, $T : S \times A \to \Delta(S)$ is a transition function, $I \in \Delta(S)$ is an initial state distribution, $\mathcal{R}$ is a reward function, the most general form of which is $\mathcal{R} : S \times A \times S \to \Delta(\mathbb{R})$, and $\gamma \in [0, 1]$ is the discount factor. Here $\Delta(X)$ is the set of all distributions over X. A **stationary policy** is a function $\pi : S \to \Delta(A)$ that specifies a distribution over actions in each state, and a **non-stationary** policy is a function $\pi : (S \times A)^* \times S \to \Delta(A)$, where $*$ is the Kleene star. A **trajectory** τ is a path $s_0, a_0, r_0, ...$ through the MDP that is possible according to T, I, and $\mathcal{R}$. The **return** of a trajectory is the discounted sum of rewards $G(\tau) \doteq \sum_{t=0}^{\infty} \gamma^t r_t$, and the **value** of a policy is the expected return $J(\pi) \doteq \mathbb{E}_{\tau\sim\pi}[G(\tau)]$. We derive **policy (preference) orderings** from reward functions by ordering policies according to their value. In this paper, we assume that S and A are finite, that $|A| > 1$, that all states are reachable, and that $\mathcal{R}(s, a, s')$ has finite mean for all s, a, s'.

In our work, we consider various reward functions for a given environment, which is then formally a **Markov decision process without reward** $MDP \setminus \mathcal{R} \doteq (S, A, T, I, _, \gamma)$. Having fixed an $MDP \setminus \mathcal{R}$, any reward function can be viewed as a function of only the current state and action by marginalizing over transitions: $\mathcal{R}(s, a) \doteq \sum_{s'\sim T(s'|s,a)} \mathcal{R}(s, a, s')$, we adopt this view from here on. We define the **(discounted) visit counts** of a policy as $\mathcal{F}^\pi(s, a) \doteq \mathbb{E}_{\tau\sim\pi}[\sum_{i=0}^{\infty} \gamma^i \mathbb{1}(s_i = s, a_i = a)]$. Note that $J(\pi) = \sum_{s,a} \mathcal{R}(s, a)\mathcal{F}^\pi(s, a)$, which we also write as $\langle \mathcal{R}, \mathcal{F}^\pi \rangle$. When considering multiple reward functions in an $MDP \setminus \mathcal{R}$, we define $J_\mathcal{R}(\pi) \doteq \langle \mathcal{R}, \mathcal{F}^\pi \rangle$ and sometimes use

[3]Their result concerns non-stationary policies and use non-Markovian reward functions, but in Appendix C, we show how an analogous construction can be used with stationary policies and Markovian rewards.

$J_i(\pi) \doteq \langle \mathcal{R}_i, \mathcal{F}^\pi \rangle$ as shorthand. We also use $\mathcal{F} : \Pi \to \mathbb{R}^{|S||A|}$ to denote the embedding of policies into Euclidean space via their visit counts, and define $\mathcal{F}(\dot{\Pi}) \doteq \{\mathcal{F}(\pi : \pi \in \dot{\Pi})\}$ for any $\dot{\Pi}$. Moreover, we also use a second way to embed policies into Euclidean space; let $\mathcal{G}(\pi)$ be the $|S||A|$-dimensional vector where $\mathcal{G}(\pi)[s, a] = \pi(a \mid s)$, and let $\mathcal{G}(\dot{\Pi}) \doteq \{\mathcal{G}(\pi : \pi \in \dot{\Pi})\}$.

4.2 Definitions and Basic Properties of Hackability and Simplification

Here, we formally define *hackability* as a binary relation between reward functions.

Definition 1. A pair of reward functions $\mathcal{R}_1$, $\mathcal{R}_2$ are **hackable** relative to policy set Π and an environment $(S, A, T, I, __, \gamma)$ if there exist $\pi, \pi' \in \Pi$ such that

$$J_1(\pi) < J_1(\pi') \;\;\&\;\; J_2(\pi) > J_2(\pi'),$$

else they are **unhackable**.

Note that an unhackable reward pair can have $J_1(\pi) < J_1(\pi') \;\;\&\;\; J_2(\pi) = J_2(\pi')$ or vice versa. Unhackability is symmetric; this can be seen be swapping π and π' in Definition 1. It is not transitive, however. In particular, the constant reward function is unhackable with respect to any other reward function, so if it *were* transitive, any pair of policies would be unhackable. Additionally, we say that $\mathcal{R}_1$ and $\mathcal{R}_2$ are **equivalent** on a set of policies Π if J_1 and J_2 induce the same ordering of Π, and that $\mathcal{R}$ is **trivial** on Π if $J(\pi) = J(\pi')$ for all $\pi, \pi' \in \Pi$. It is clear that $\mathcal{R}_1$ and $\mathcal{R}_2$ are unhackable whenever they are equivalent, or one of them is trivial, but this is relatively uninteresting. Our central question is if and when there are other unhackable reward pairs.

The symmetric nature of this definition is counter-intuitive, given that our motivation distinguishes the proxy and true reward functions. We might break this symmetry by only considering policy sequences that monotonically increase the proxy, however, this is equivalent to our original definition of hackability: think of $\mathcal{R}_1$ as the proxy, and consider the sequence π, π'. We could also restrict ourselves to policies that are approximately optimal according to the proxy; Corollary 2 shows that Theorem 1 applies regardless of this restriction. Finally, we define *simplification* as an asymmetric special-case of unhackability; Theorem 3 shows this is in fact a more demanding condition.

Definition 2. $\mathcal{R}_2$ is a **simplification** of $\mathcal{R}_1$ relative to policy set Π if for all $\pi, \pi' \in \Pi$,

$$J_1(\pi) < J_1(\pi') \implies J_2(\pi) \leq J_2(\pi') \;\;\&\;\; J_1(\pi) = J_1(\pi') \implies J_2(\pi) = J_2(\pi')$$

and there exist $\pi, \pi' \in \Pi$ such that $J_2(\pi) = J_2(\pi')$ but $J_1(\pi) \neq J_1(\pi')$. Moreover, if $\mathcal{R}_2$ is trivial then we say that this is a **trivial simplification**.

Intuitively, while unhackability allows replacing inequality with equality – or vice versa – a simplification can only replace inequalities with equality, collapsing distinctions between policies. When $\mathcal{R}_1$ is a simplification of $\mathcal{R}_2$, we also say that $\mathcal{R}_2$ is a **refinement** of $\mathcal{R}_1$. We denote this relationship as $\mathcal{R}_1 \trianglelefteq \mathcal{R}_2$ or $\mathcal{R}_2 \trianglerighteq \mathcal{R}_1$; the narrowing of the triangle at R_1 represents the collapsing of distinctions between policies. If $\mathcal{R}_1 \trianglelefteq \mathcal{R}_2 \trianglerighteq \mathcal{R}_3$, then we have that $\mathcal{R}_1, \mathcal{R}_3$ are unhackable,[4] but if $\mathcal{R}_1 \trianglerighteq \mathcal{R}_2 \trianglelefteq \mathcal{R}_3$, then this is not necessarily the case.[5]

Note that these definitions are given relative to some $MDP \setminus \mathcal{R}$, although we often assume the environment in question is clear from context and suppress this dependence. The dependence on the policy set Π, on the other hand, plays a critical role in our results.

5 Results

Our results are aimed at understanding when it is possible to have an unhackable proxy reward function. We first establish (in Section 5.1) that (non-trivial) unhackability is impossible when considering the set of all policies. We might imagine that restricting ourselves to a set of sufficiently good (according to the proxy) policies would remove this limitation, but we show that this is not the case. We then analyze finite policy sets (with deterministic policies as a special case), and establish necessary and sufficient conditions for unhackability and simplification. Finally, we demonstrate via example that non-trivial simplifications are also possible for some infinite policy sets in Section 5.3.

[4] If $J_3(\pi) > J_3(\pi')$ then $J_2(\pi) > J_2(\pi')$, since $\mathcal{R}_2 \trianglerighteq \mathcal{R}_3$, and if $J_2(\pi) > J_2(\pi')$ then $J_1(\pi) \geq J_1(\pi')$, since $\mathcal{R}_1 \trianglelefteq \mathcal{R}_2$. It is therefore not possible that $J_3(\pi) > J_3(\pi')$ but $J_1(\pi) < J_1(\pi')$.

[5] Consider the case where $\mathcal{R}_2$ is trivial – then $\mathcal{R}_1 \trianglerighteq \mathcal{R}_2 \trianglelefteq \mathcal{R}_3$ for any $\mathcal{R}_1, \mathcal{R}_3$.

5.1 Non-trivial Unhackability Requires Restricting the Policy Set

We start with a motivating example. Consider the setting shown in Figure 3, where the agent can move left/stay-still/right and gets a reward depending on its state. Let the Gaussian (blue) be the true reward $\mathcal{R}_1$ and the step function (orange) be the proxy $\mathcal{R}_2$. These are hackable. To see this, consider being at state B. Let $\pi(B)$ travel to A or C with 50/50 chance, and compare with the policy π' that stays at B. Then we have that $J_1(\pi) > J_1(\pi')$ and $J_2(\pi) < J_2(\pi')$.

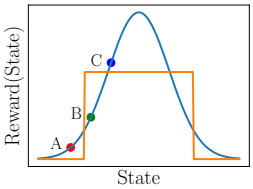

Figure 3: Two reward functions. While the step function may seem like a simplification of the Gaussian, these reward functions are hackable.

Generally, we might hope that some environments allow for unhackable reward pairs that are not equivalent or trivial. Here we show that this is not the case, unless we impose restrictions on the set of policies we consider.

First note that if we consider *non-stationary* policies, this result is relatively straightforward. Suppose $\mathcal{R}_1$ and $\mathcal{R}_2$ are *unhackable* and *non-trivial* on the set Π^N of all non-stationary policies, and let $\pi^\star$ be a policy that maximises ($\mathcal{R}_1$ and $\mathcal{R}_2$) reward, and $\pi_\perp$ be a policy that *minimises* ($\mathcal{R}_1$ and $\mathcal{R}_2$) reward. Then the policy π_λ that plays $\pi^\star$ with probability λ and $\pi_\perp$ with probability $1-\lambda$ is a policy in Π^N. Moreover, for any π there are two unique $\alpha, \beta \in [0,1]$ such that $J_1(\pi) = J_1(\pi_\alpha)$ and $J_2(\pi) = J_2(\pi_\beta)$. Now, if $\alpha \neq \beta$, then either $J_1(\pi) < J_1(\pi_\delta)$ and $J_2(\pi) > J_2(\pi_\delta)$, or vice versa, for $\delta = (\alpha+\beta)/2$. If $\mathcal{R}_1$ and $\mathcal{R}_2$ are unhackable then this cannot happen, so it must be that $\alpha = \beta$. This, in turn, implies that $J_1(\pi) = J_1(\pi')$ iff $J_2(\pi) = J_2(\pi')$, and so $\mathcal{R}_1$ and $\mathcal{R}_2$ are *equivalent*. This means that no interesting unhackability can occur on the set of all non-stationary policies.

The same argument cannot be applied to the set of *stationary* policies, because π_λ is typically not stationary, and mixing stationary policies' action probabilities does not have the same effect. For instance, consider a hallway environment where an agent can either move left or right. Mixing the "always go left" and "always go right" policies corresponds to picking a direction and sticking with it, whereas mixing their action probabilities corresponds to choosing to go left or right independently at every time-step. However, we will see that there still cannot be any interesting unhackability on this policy set, and, more generally, that there cannot be any interesting unhackability on any set of policies which contains an *open subset*. Formally, a set of (stationary) policies $\dot{\Pi}$ is open if $\mathcal{G}(\dot{\Pi})$ is open in the smallest affine space that contains $\mathcal{G}(\Pi)$, for the set of all stationary policies Π. We will use the following lemma:

Lemma 1. *In any $MDP \setminus \mathcal{R}$, if $\dot{\Pi}$ is an open set of policies, then $\mathcal{F}(\dot{\Pi})$ is open in $\mathbb{R}^{|S|(|A|-1)}$, and $\mathcal{F}$ is a homeomorphism between $\mathcal{G}(\dot{\Pi})$ and $\mathcal{F}(\dot{\Pi})$.*

Using this lemma, we can show that interesting unhackability is impossible on any set of stationary policies $\hat{\Pi}$ which contains an open subset $\dot{\Pi}$. Roughly, if $\mathcal{F}(\dot{\Pi})$ is open, and $\mathcal{R}_1$ and $\mathcal{R}_2$ are non-trivial and unhackable on $\dot{\Pi}$, then the fact that J_1 and J_2 have a linear structure on $\mathcal{F}(\hat{\Pi})$ implies that $\mathcal{R}_1$ and $\mathcal{R}_2$ must be equivalent on $\dot{\Pi}$. From this, and the fact that $\mathcal{F}(\dot{\Pi})$ is open, it follows that $\mathcal{R}_1$ and $\mathcal{R}_2$ are equivalent everywhere.

Theorem 1. *In any $MDP \setminus \mathcal{R}$, if $\hat{\Pi}$ contains an open set, then any pair of reward functions that are unhackable and non-trivial on $\hat{\Pi}$ are equivalent on $\hat{\Pi}$.*

Since simplification is a special case of unhackability, this also implies that non-trivial simplification is impossible for any such policy set. Also note that Theorem 1 makes *no assumptions* about the transition function, etc. From this result, we can show that interesting unhackability always is impossible on the set Π of all (stationary) policies. In particular, note that the set $\tilde{\Pi}$ of all policies that always take each action with positive probability is an open set, and that $\tilde{\Pi} \subset \Pi$.

Corollary 1. *In any $MDP \setminus \mathcal{R}$, any pair of reward functions that are unhackable and non-trivial on the set of all (stationary) policies Π are equivalent on Π.*

Theorem 1 can also be applied to many other policy sets. For example, we might not care about the hackability resulting from policies with low proxy reward, as we would not expect a sufficiently good learning algorithm to learn such policies. This leads us to consider the following definition: